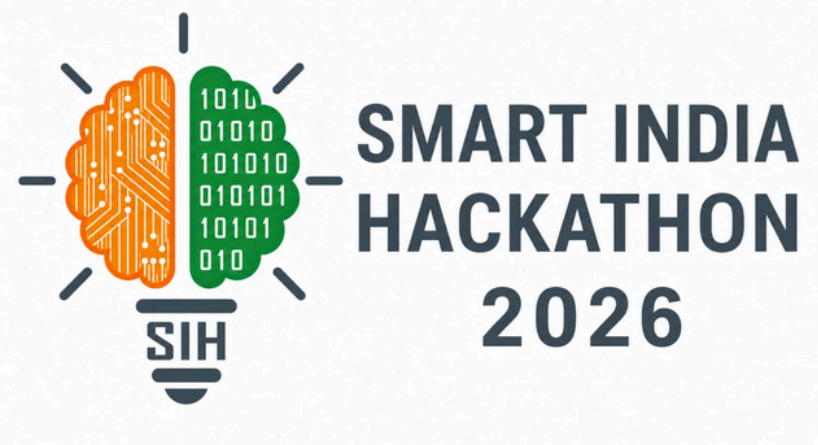




SMART INDIA HACKATHON 2026

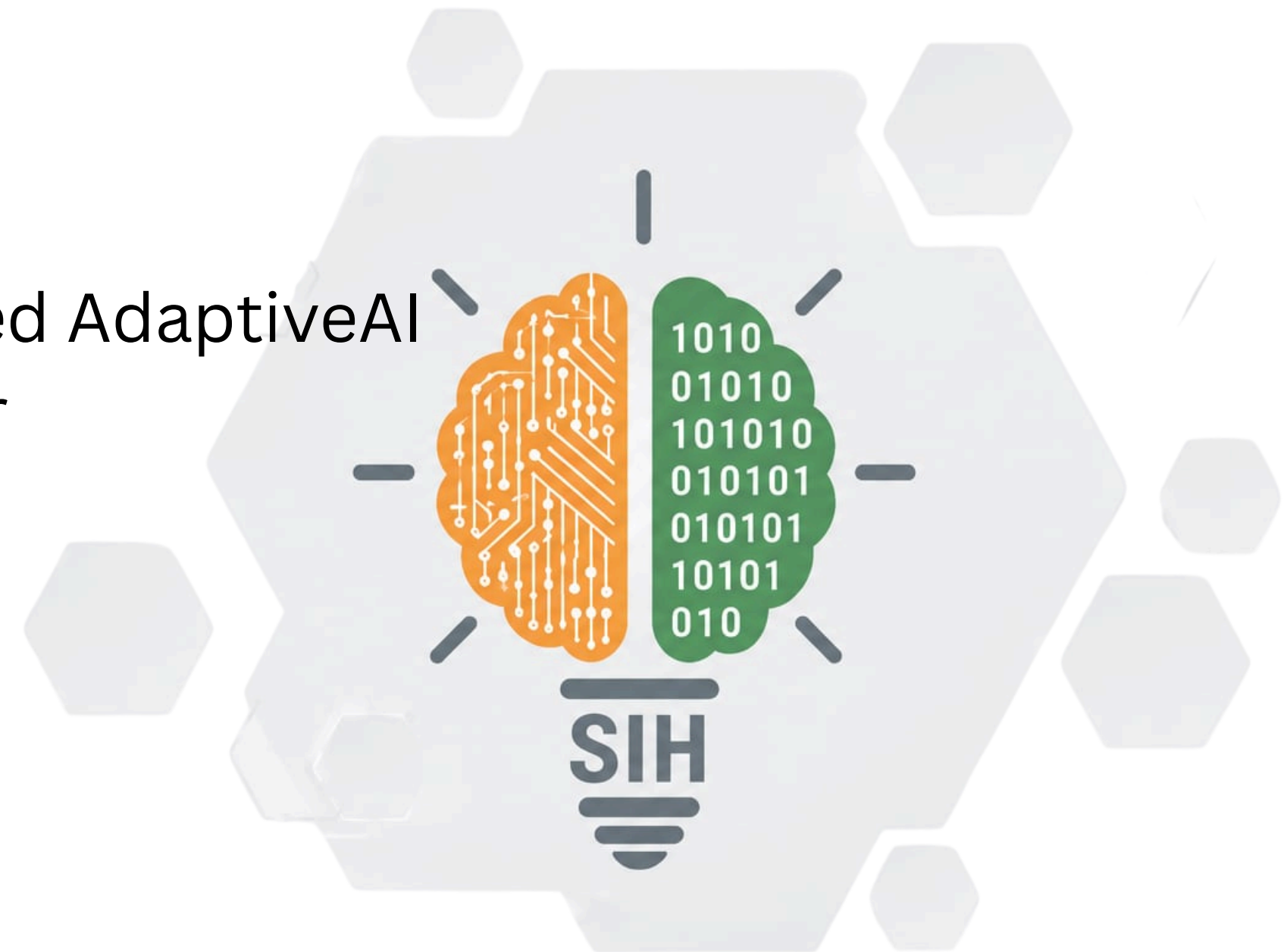
Problem Statement ID - SOAIDEATHON-S28

Problem Statement Title - Curriculum-Aligned AdaptiveAI
Tutor

Theme - Smart Automation

PS Category - Software

Team Name - NOVA NEXUS



concepta

CONCEPTA

Curriculum-Grounded Learning Co-Pilot

 **The Problem**

AI CAN ANSWER.
STUDENTS STILL MAY NOT UNDERSTAND.

Generative AI often responds to the question asked without identifying the underlying learning gap. A student struggling with recursion, for example, may actually be missing prerequisites such as functions, call stack, or base case.



The real problem isn't always the question.
It's the **missing concept** behind it.

 OUR SOLUTION

**CURRICULUM-ALIGNED
ADAPTIVE AI TUTOR**

Our system builds a student learning profile, identifies prerequisite gaps, and uses institution-approved curriculum content to deliver explanations and practice adapted to the learner's demonstrated understanding.



CURRICULUM-GROUNDED

Uses approved course content



GAP-AWARE

Finds missing prerequisites



ADAPTIVE

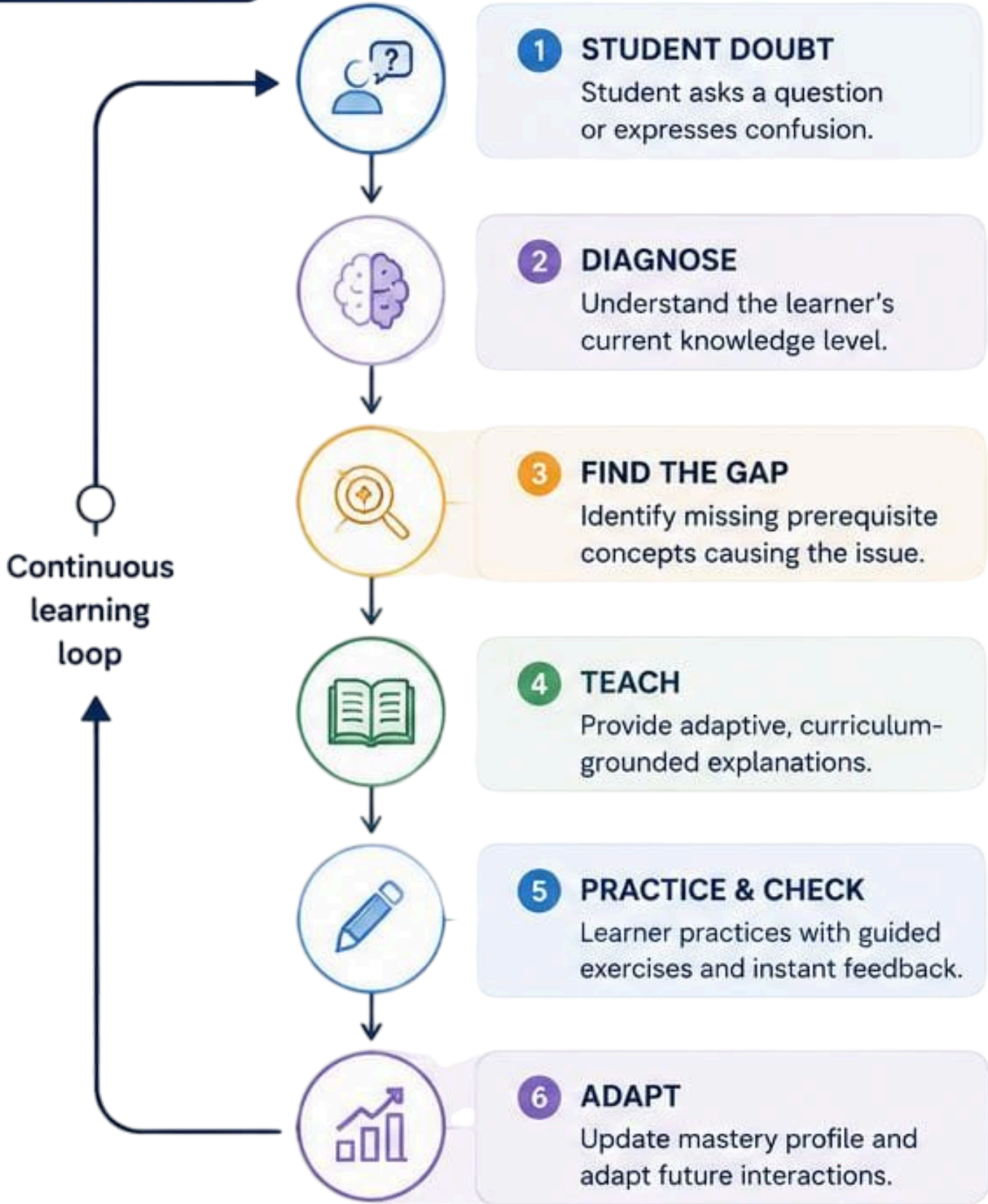
Adjusts explanation & difficulty



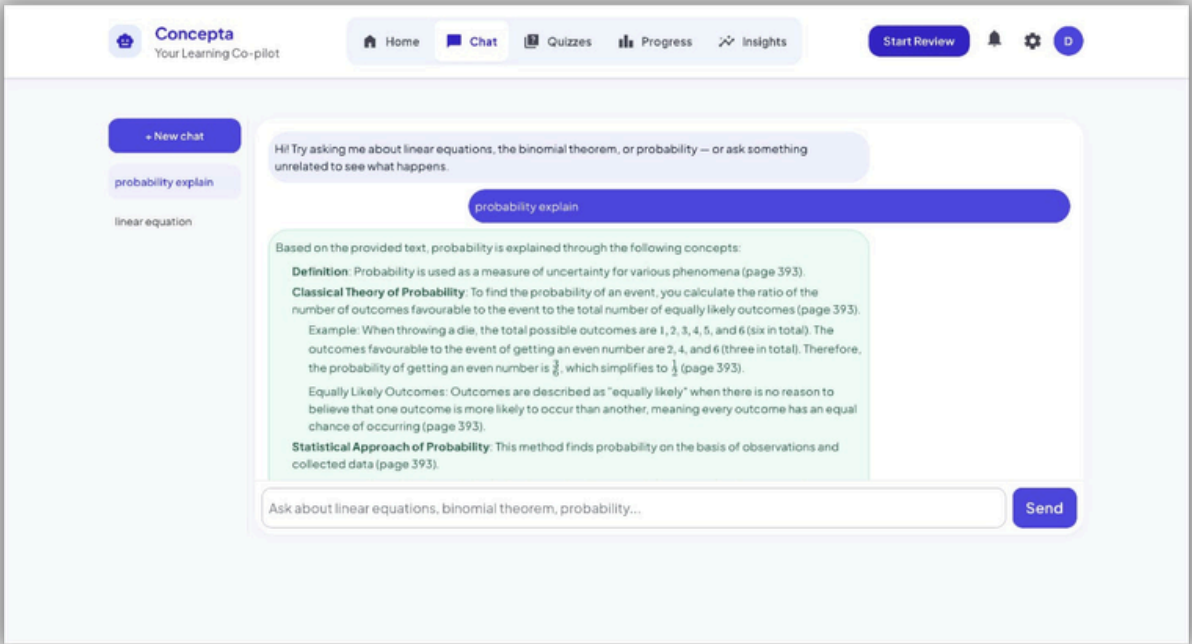
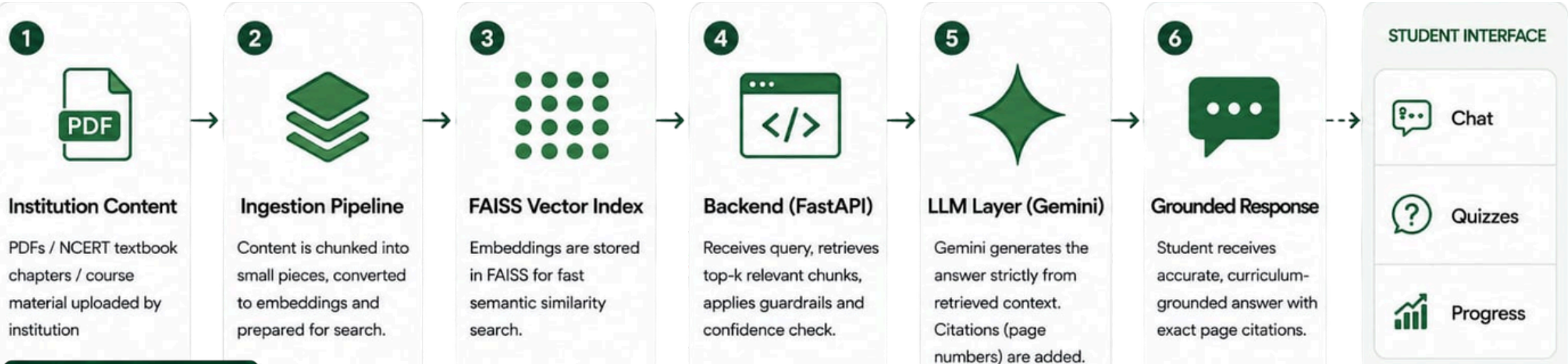
EVIDENCE-GROUNDED

Answers backed by retrieved sources

HOW IT WORKS



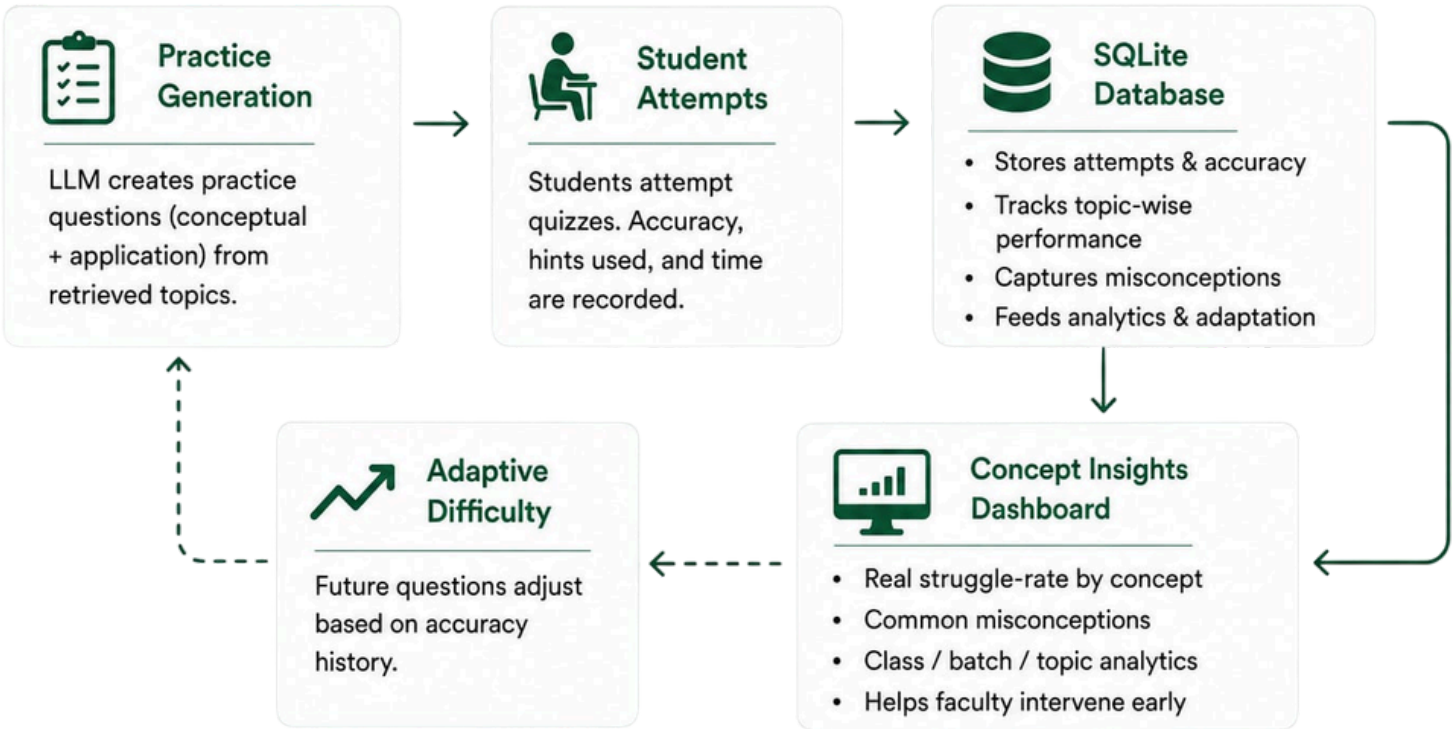
SYSTEM PIPELINE



TECH STACK

	Frontend Framework	React (via Vite)
	Styling	Tailwind CSS + Material Symbols icons
	Charts	Recharts
	State Management	React state / Context API
	Communication	Fetch API (REST calls to backend)
	Backend Technology	Python, FastAPI, Uvicorn (ASGI server) FastAPI CORS middleware
	Embeddings Model	sentence-transformers (all-MiniLM-L6-v2)
	Vector Search Library	FAISS (semantic similarity search)
	LLM	Google Gemini (via google-generativeai)
	Database	SQLite (practice attempts + insights)
	Cross-origin Handling	FastAPI CORS middleware

DATA FLOW & FEEDBACK LOOP



Confidence-Threshold Guardrail: If confidence is low, the system asks a clarifying question instead of guessing — ensuring reliability.

WHY THIS APPROACH WORKS



Grounded & Curriculum-Locked
Answers come only from institution-uploaded material — no open web, no hallucinations.



Verifiable & Transparent
Every answer is backed by exact page citation from the source.



Adaptive & Data-Driven
Difficulty adapts using real accuracy history; insights highlight true learning gaps.

WHO BENEFITS?



Students Get accurate answers, practice, and track progress.



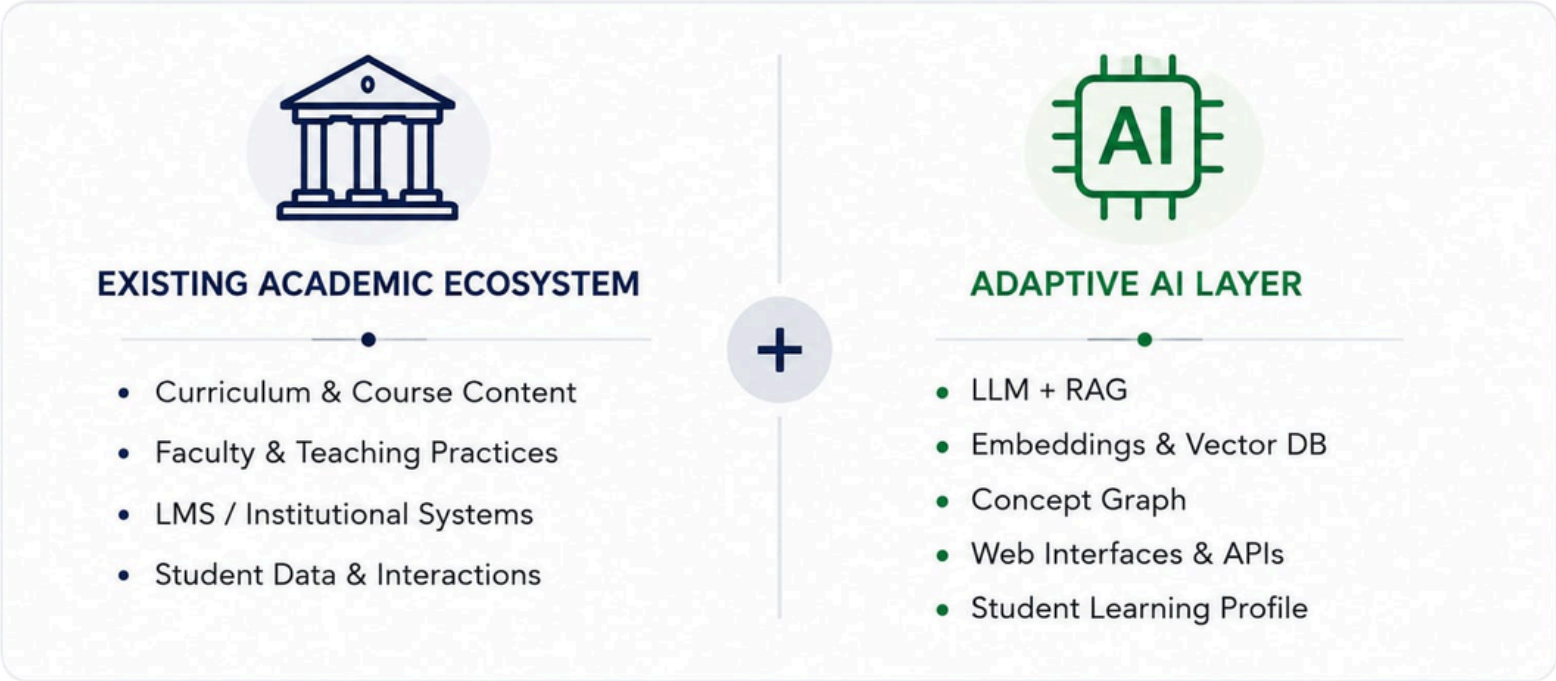
Faculty See real conceptual gaps and class-level insights.



Institutions Maintain complete control over curriculum content and quality.

01 BUILT AROUND EXISTING SYSTEMS

Works alongside current academic ecosystem.



✓ We **enhance** what exists, not replace it.

02 PILOT → SCALE

A phased, practical, and scalable deployment strategy.



03 HOW WE PROVE IT WORKS

Clear metrics. Real impact.



Our goal is not more AI-generate answers, but better student understanding through personalised learning. Targeted practice and continuous feedback.

EXISTING SOLUTIONS (GAPS)	VS.	OUR APPROACH (NOVA NEXUS)
 Generic AI Knowledge Responses are not grounded in institution-approved content.	➔	 Curriculum-Grounded Support Works with institution-approved content and teaching practices.
 No Source-Level Verification Does not provide exact page-level citations or source verification.	➔	 Verifiable & Traceable Answers Provides exact source and page-level citations for every answer.
 Limited Misconception Tracking Lacks mechanisms to identify and track student misconceptions.	➔	 Learning Intelligence Identifies common misconceptions and learning gaps through data insights.
 No Adaptive Practice Same difficulty for all, with no adaptation based on accuracy history.	➔	 Adaptive Practice Generation Generates personalised practice based on student accuracy and concept mastery.



STUDENTS



Personalised Learning

Explanation and practice adapt to the learner's demonstrated understanding and pace.



Learning-Gap Identification

Identifies missing prerequisite concepts and addresses the root cause of confusion.



Faster, Targeted Doubt Identification

Students receive help focused on where they are actually stuck, followed by practice and feedback.



FACULTY



Misconception Insights

See common misconceptions and learning gaps across students through data-driven insights.



Reduces Repetitive Doubt-Solving

The tutor handles routine academic support so faculty can focus on higher-value teaching.



Focus on High-Value Teaching

Enables faculty to spend more time on conceptual depth, mentoring and personalised guidance.



INSTITUTIONS



Curriculum-Aligned Support

Works with institution-approved content and teaching practices, not generic AI knowledge.



Scalable Academic Support

Can expand from a pilot to more courses, departments and institutions while staying aligned with curriculum.



Verifiable, Reliable and Accountable

Provides verifiable, reliable and accountable AI support for academic excellence.



ACADEMIC & TECHNICAL REFERENCES

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<https://arxiv.org/abs/2005.11401>
- 2** Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 3982–3992.
<https://arxiv.org/abs/1908.10084>
- 3** Johnson, J., Douze, M., & Jégou, H. (2019). Billion-Scale Similarity Search with GPUs. *IEEE Transactions on Big Data*, 7(3), 535–547.
<https://doi.org/10.1109/TBDATA.2019.2921572>
- 4** Gemini Team, Google. (2023). Gemini: A Family of Highly Capable Multimodal Models.
<https://arxiv.org/abs/2312.11805>

WHERE CONCEPTA DIFFERS (GAP IDENTIFIED)

- | | |
|---|--|
| ✓ Answers are generated only from institution-uploaded materials. | ✓ Adaptive difficulty adjusts practice based on accuracy history. |
| ✓ Every response includes exact page-level citation for verification. | ✓ Provides concept-level analytics for both students and faculty. |
| ✓ Concept Insights Dashboard tracks misconceptions and learning gaps. | ✓ Ensures a transparent, curriculum-locked, and data-driven learning experience. |

RESEARCH REVIEW



We reviewed current AI tutoring tools and learning platforms to identify gaps in providing curriculum-grounded, verifiable, and adaptive learning support.

EXISTING SOLUTIONS REVIEWED

- **Khanmigo (Khan Academy)**
 - Uses GPT-4 to provide hints and explanations through Socratic questioning.
 - Strongly tied to Khan Academy’s own content and exercises.
 - Does not provide exact page-level citations from source materials.
 - Limited support for tracking misconceptions or adaptive difficulty based on accuracy history.
- **Generic AI Chatbots (ChatGPT, Gemini, etc.)**
 - Provide broad, web-based knowledge and general explanations.
 - Responses are not grounded in institution-uploaded or curriculum-locked material.
 - Do not provide exact source-page citations.
 - Lack mechanisms for misconception tracking and concept-level analytics.
 - No adaptive practice generation based on real accuracy history.